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13. Running WordPress & Database Containers on Portenta X8

Learn how to run a database and WordPress container on the Portenta X8

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Overview

The Arduino Portenta X8's robust features are ideally complemented by Docker containers, simplifying various applications. This tutorial demonstrates how to deploy a WordPress web server on the Portenta X8, leveraging containers for web service and database management.

You will learn to set up and access a WordPress site hosted on the X8 via a web browser.

Goals

- Prepare the necessary files for Docker container deployment
- Deploy and activate the Docker containers on the Portenta X8
- Access and configure the WordPress site hosted on the Portenta X8

Required Hardware and Software

Portenta X8

USB-C® cable (USB-C® to USB-A cable)

The `docker-compose.yml` file used in this tutorial

Instructions

Begin by ensuring your Portenta X8 is ready for use, following the setup guide in the [User Manual's Out-of-the-Box Experience](#).

Creating the `docker-compose.yml` File

Our WordPress setup involves a multi-container approach, integrating both the WordPress and a MariaDB database server containers. The WordPress container uses Apache as the web server, merged within the container for easy deployment.

To direct this setup, we will craft a `docker-compose.yml` file describing the configurations for both WordPress and MariaDB containers, including essential settings like usernames, passwords, time zones, and database names. For security, ensure the default passwords are substituted with stronger alternatives in the provided configuration template.

Complete `docker-compose.yml` File

In this section, you can find the complete **docker-compose.yml** file that we will be using for this tutorial.

```
1  version: "3.9"
2
3  services:
4    db:
5      image: mariadb:late
6      container_name: mar
7      environment:
8        - PUID=1000
9        - PGID=1000
10       - MYSQL_ROOT_PASS
11       - TZ=Europe/Londo
12       - MYSQL_DATABASE=
13       - MYSQL_USER=Word
14       - MYSQL_PASSWORD=
15     volumes:
16       - db_data:/var/li
17     restart: unless-sto
18
19   Wordpress:
20     depends_on:
21       - db
22     image: wordpress:la
23     volumes:
24       - Wordpress_data:
25     ports:
26       - "8000:80"
27     restart: always
28     environment:
```

Now, let's prepare our Portenta X8 by creating a directory for our **docker-compose.yml** file, which can be downloaded [here](#).

Installing The Containers

Begin by creating a directory for the Docker setup, naming it **wordpress-test**. Navigate into this directory and either copy the *docker-compose.yml* file into it or create the file directly within.

To create the file on the device, use

the contents, and exit with

`CTRL + C`. To transfer the file from your computer, use the following command, making sure to replace

```
<path to docker-compose.yml  
file>
```

with the actual file path:

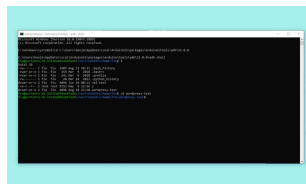
```
1 adb push <path to docker-
```

Alternatively, you could place the `docker-compose.yml` file inside the `wordpress-test` directory and push the file using the following command:

Alternatively, if the `docker-compose.yml` is already inside the **wordpress-test** directory on your computer, use:

```
1 adb push .\wordpress-test
```

Choose the method that best suits your workflow.



cd into correct directory



Access Docker with administrative privileges by executing `sudo su -`, with `fio` as the default password.

Ensure no conflicting containers are

`docker ps -a`. Remove any active containers by first stopping them with:

```
1 docker stop <container id>
```

Then removing them with:

```
1 docker rm <container id>
```

If you want more information about handling containers on your Portenta X8, take a look at our [Managing Containers with Docker tutorial](#).

With the setup directory ready and no port conflicts, begin the container installation with:

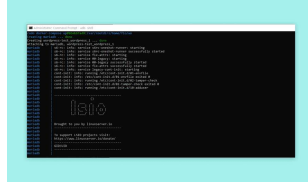
```
1 docker compose up -d
```

The `-d` flag runs the containers in the background; omitting it will tie the container's lifecycle to the terminal session.

The installation of the **WordPress** and **MariaDB** containers will begin and may take some time. To monitor the installation process, use:

```
1 docker compose logs -f
```

Portenta X8.



Containers install
progress in the terminal

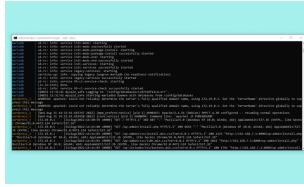
Connecting to the WordPress Site

Accessing your WordPress site on the Portenta X8 is straightforward. Use the following URL format, composed with your Portenta X8's unique id and port, in your browser:

```
1 http://portenta-x8-<uuid>
```

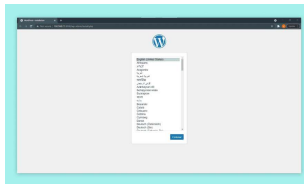
Replace `<uuid>` with your Portenta X8's unique identifier and `<port>` with the port you have allocated for the WordPress container. You can find your device's `<uuid>` in the setup guide within the [User Manual's Out-of-the-Box Experience](#), through terminal commands involving `adb`, or by visiting `http://192.168.7.1:8000` on Windows and Linux (use `http://192.168.8.1:8000` for MacOS).

Upon establishing a connection, your terminal will display details similar to those below.



Terminal printout during connection

Your browser will then present the WordPress setup page, allowing you to commence the configuration process.



Wordpress setup site

You are now free to go through the WordPress setup process and configure it however you like.

Removing the Containers

Should you need to remove the containers, navigate back to the `/home/fio/wordpress-test` directory and use the commands below based on your requirements.

To remove the containers while retaining your WordPress data:

```
1 docker compose down
```

To delete both the containers and all associated data:

Confirm the removal by executing `docker ps -a` and verifying that the WordPress and MariaDB containers are no longer listed.

Conclusion

In this tutorial, we went through installing and running a WordPress and database container on the Portenta X8. We then accessed the WordPress site on our X8 through our web browser. Now, you can set up your WordPress site on your X8 device and access it from another device.

Troubleshooting

If the containers are not being installed or running correctly, check if any other containers are currently running on the same ports as the ones used by the WordPress container. You can check it with

```
docker ps -a .
```

If there is any issue running docker commands, ensure you are using `sudo` before the commands or having root access using: `sudo su -r` with the password you selected at first access.

If you cannot connect to the site when everything is running, you can double-check the X8's IP address. Run the command `ip -h address` in the **adb shell**. This will display the X8's IP address via USB

those IP addresses if all the rest fails.

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